

Aerodynamic Principles of Subsonic Airfoils

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Abstract & Governing Equations

This document establishes the governing aerodynamic relations for subsonic wing design.

For incompressible flow, the lift coefficient is defined as:

$$C_L = 2 * L / (\rho * v^2 * S)$$

where:

$L = 12500.0$ N (aerodynamic lift force)

$\rho = 1.225$ kg/m³ (air density at sea level)

$v = 68.0$ m/s (free-stream flight velocity)

$S = 16.2$ m² (wing reference surface area)

The lift coefficient C_L is an essential non-dimensional figure of merit for aeronautical performance.

Section 2: Drag Polar & Induced Drag

The total drag coefficient C_D consists of profile drag C_{D0} and induced drag C_{Di} :

$$C_D = C_{D0} + (C_L^2) / (\pi * e * AR)$$

where:

$C_{D0} = 0.0210$ (zero-lift parasite drag coefficient)

$e = 0.85$ (Oswald span efficiency factor)

$AR = 8.4$ (wing aspect ratio)

The lift-to-drag ratio L/D determines range efficiency and aerodynamic cruising economy.

Section 3: Boundary Layer Separation & Stall

Boundary layer separation occurs when adverse pressure gradients overcome kinetic energy in the boundary layer.
For the NACA 64-416 airfoil, aerodynamic stall occurs at a critical angle of attack $\alpha_{\text{stall}} = 14.2$ degrees.
Under standard cruise conditions at altitude, the operating Reynolds number is $Re = 6.5e6$ at Mach number $M = 0.78$.
Boundary layer thickness at the mid-chord transition is measured at $\delta = 0.012$ m with peak vorticity $\omega = 42.8 \text{ s}^{-1}$.